21 October 2026

The Editors-in-Chief
Advances in Structural Engineering
Sage

Dear Editors,

Submission of an original research article: “Blind spots and sensor placement for influence-area weigh-in-motion on continuous girders”

We submit the above manuscript for consideration as an Original Research Article in *Advances in Structural Engineering*.

Influence-area weighing divides the area under one bridge response record by an integral constant to estimate the gross weight. On continuous girders this constant can vanish, because influence lines change sign over the piers, and engineers have had no criterion for telling, before installation, which channels and positions can weigh.

The manuscript gives such criteria. Closed forms locate the blind points of five channel types: strain gauges that measure global bending are blind at every zero of the uniform-load moment and nearly blind at 0.19–0.32 of the span from the interior support on two equal spans. For position-independent noise, optimal sensors lie at the extremes of the uniform-load response. The constant follows a linear-fractional law in pier stiffness; on an in-service 55 + 90 + 55 m box girder, two static readings leave the side-span deflection constant of the base model uncertain by a factor of 6.6 against 1.5 at mid-span of the main span (2.4 against 1.1 with girder rigidity bounded to 0.9–1.3 times the design value).

The work fits the journal’s focus on structural health monitoring for performance assessment of bridges: it tells engineers where to place the few channels that screen continuous bridges for overloads.

A laboratory test is consistent with the identity. The field records are used to check the assumptions behind the criteria; they cannot test weight discrimination, because their reference weights vary little, and the manuscript states this. The moving-load test of the in-service bridge is designed but not yet performed, and its predicted outcomes are given so that they can be checked.

The manuscript is just under 12 000 word-equivalents with 7 figures, 3 numbered tables and a table of symbols; derivations, full field tables and the error budget are in the Supplementary Material.

Data availability. The field strain records are from the public NOR-BWIM dataset (figshare, https://doi.org/10.7294/26880898.v1, CC0 license). The laboratory data and the processed data of the in-service bridge are tabulated in the Supplementary Material; these data, the derived data behind the figures and tables, and all analysis scripts with their console output are archived on Zenodo (https://doi.org/10.5281/zenodo.23535533, CC BY 4.0).

The manuscript is original, has not been published, and is not under consideration by any other journal. All authors have approved the submission. Conflicts of interest, funding and the use of generative AI are declared in the manuscript.

Sincerely,

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